

FURTHER MECHANICS

Mock Paper 2 (Set B)

Time Allowed: 1 hour 30 minutes • Total Marks: 75

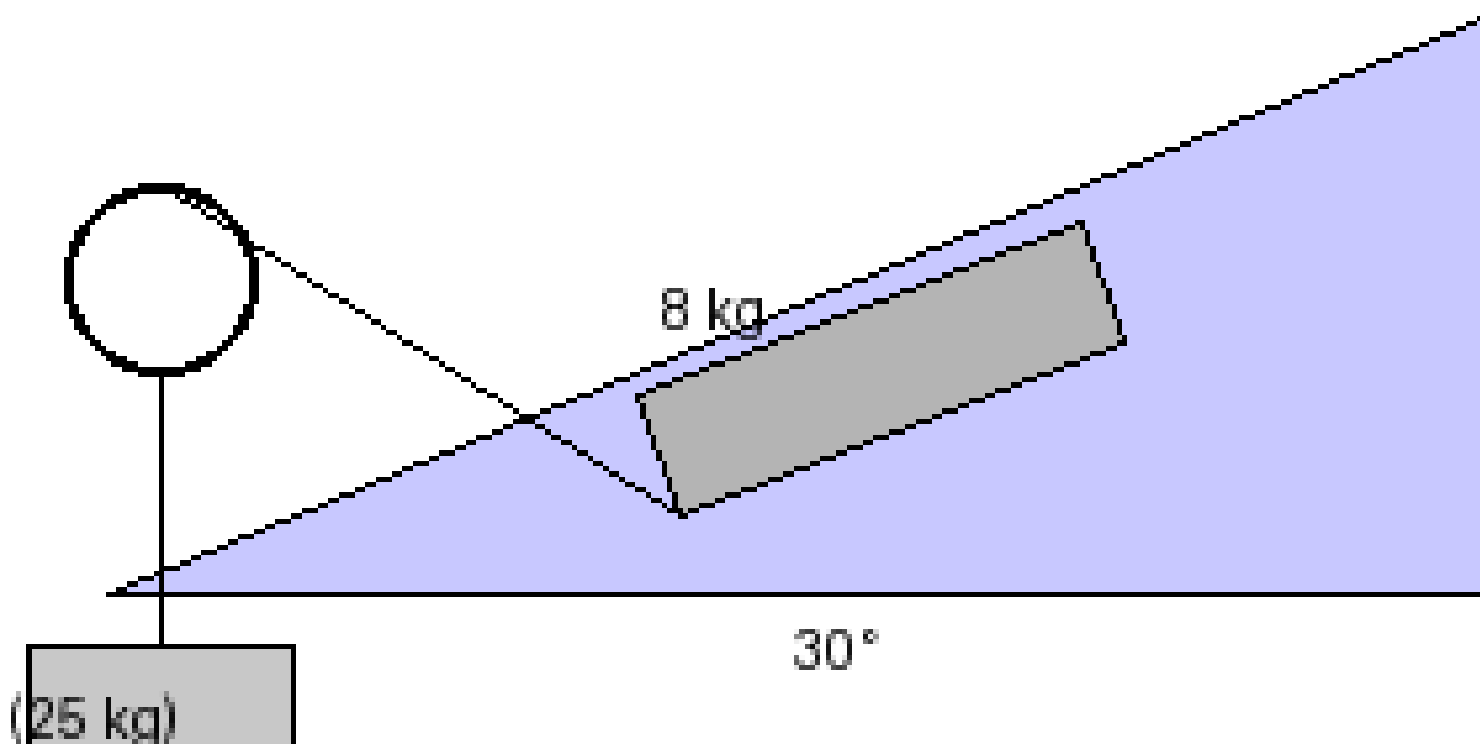
Candidate Name _____ Candidate Number _____

Instructions

- Use black or dark blue pen. Pencil may be used for diagrams.
- Answer **all** questions.
- Unless otherwise stated, take $g = 9.8 \text{ m s}^{-2}$ and give answers to an appropriate degree of accuracy.
- Non-exact numerical answers should be given to 3 significant figures unless stated otherwise.
- The total number of marks for this paper is 75.
- You are permitted to use a scientific calculator.

1. [11 marks]

A light inextensible string passes over a small smooth pulley fixed at the top of a smooth plane inclined at 30° to the horizontal. An 8 kg block lies on the plane and is attached to one end of the string. A particle of mass M kg hangs vertically below the pulley at the other end. The system is shown in the diagram below.

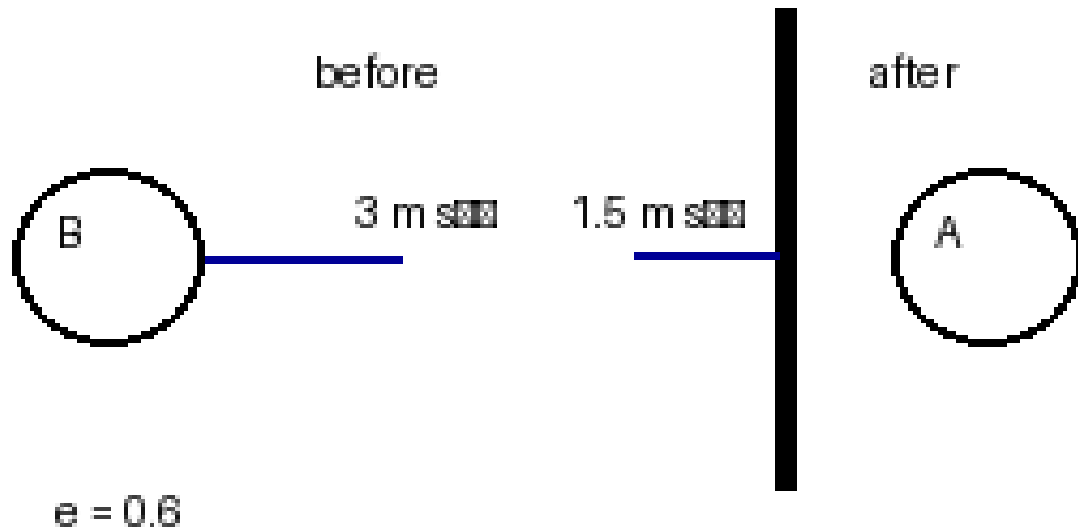


The system is released from rest with the string taut.

- Find, in terms of M , the acceleration of the 8 kg block up the plane. [3]
- Given that the acceleration of the block is 2.1 m s^{-2} , find the value of M . [3]
- Find the tension in the string. [2]
- The string is now cut. State, with a reason, whether the 8 kg block will begin to slide back down the plane. [3]

2. [8 marks]

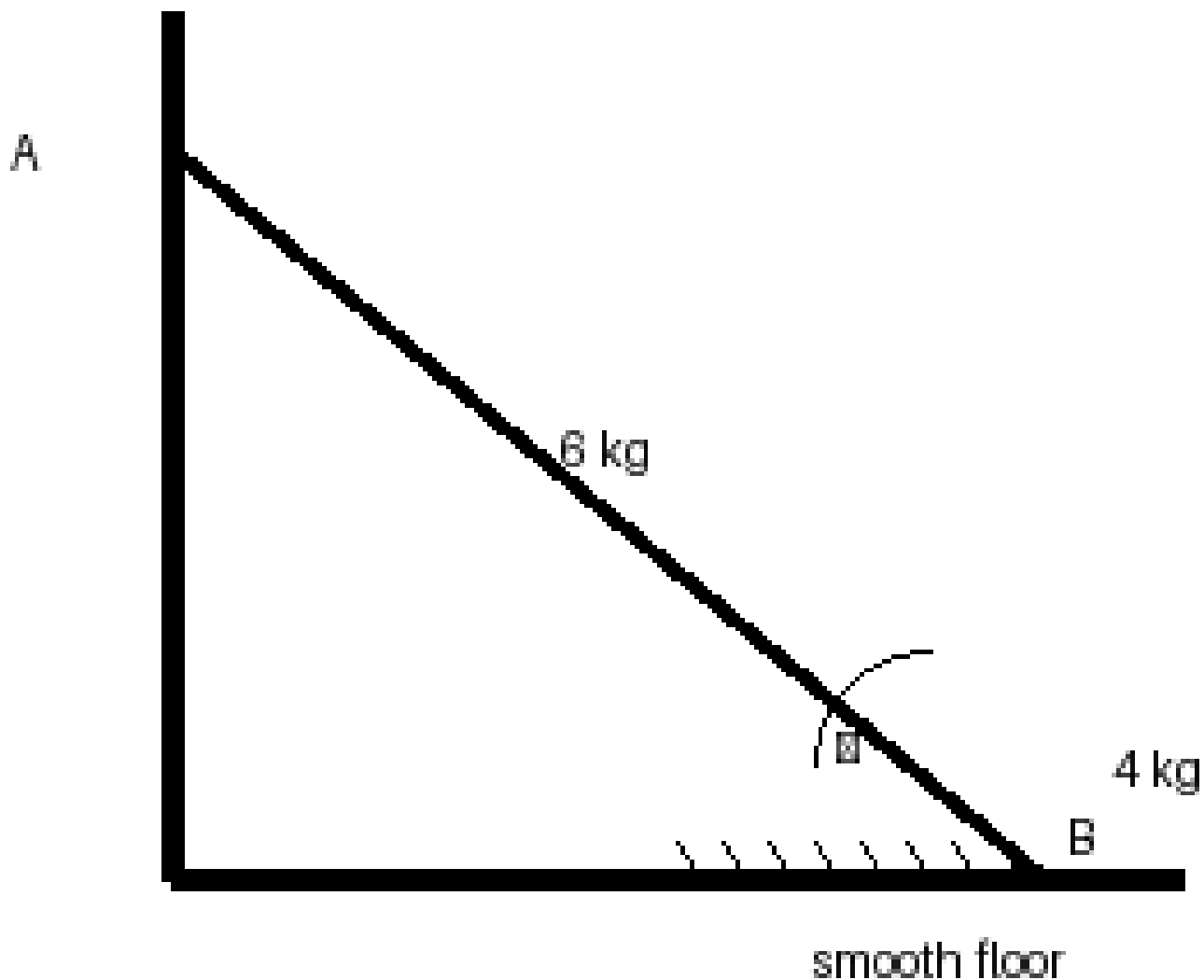
A particle A of mass 2 kg moving with speed 3 m s^{-1} directly towards a smooth vertical wall collides with the wall. The coefficient of restitution between the particle and the wall is 0.6. The situation is shown in the diagram below.



- (a) Find the speed of the particle after the impact. [2]
 (b) Find the impulse exerted by the wall on the particle, stating its direction. [3]
 (c) A second particle *B* of mass 3 kg approaches the same wall with speed 4 m s⁻¹ and rebounds with speed 2 m s⁻¹. Find the coefficient of restitution for this second collision. [3]

3. [12 marks]

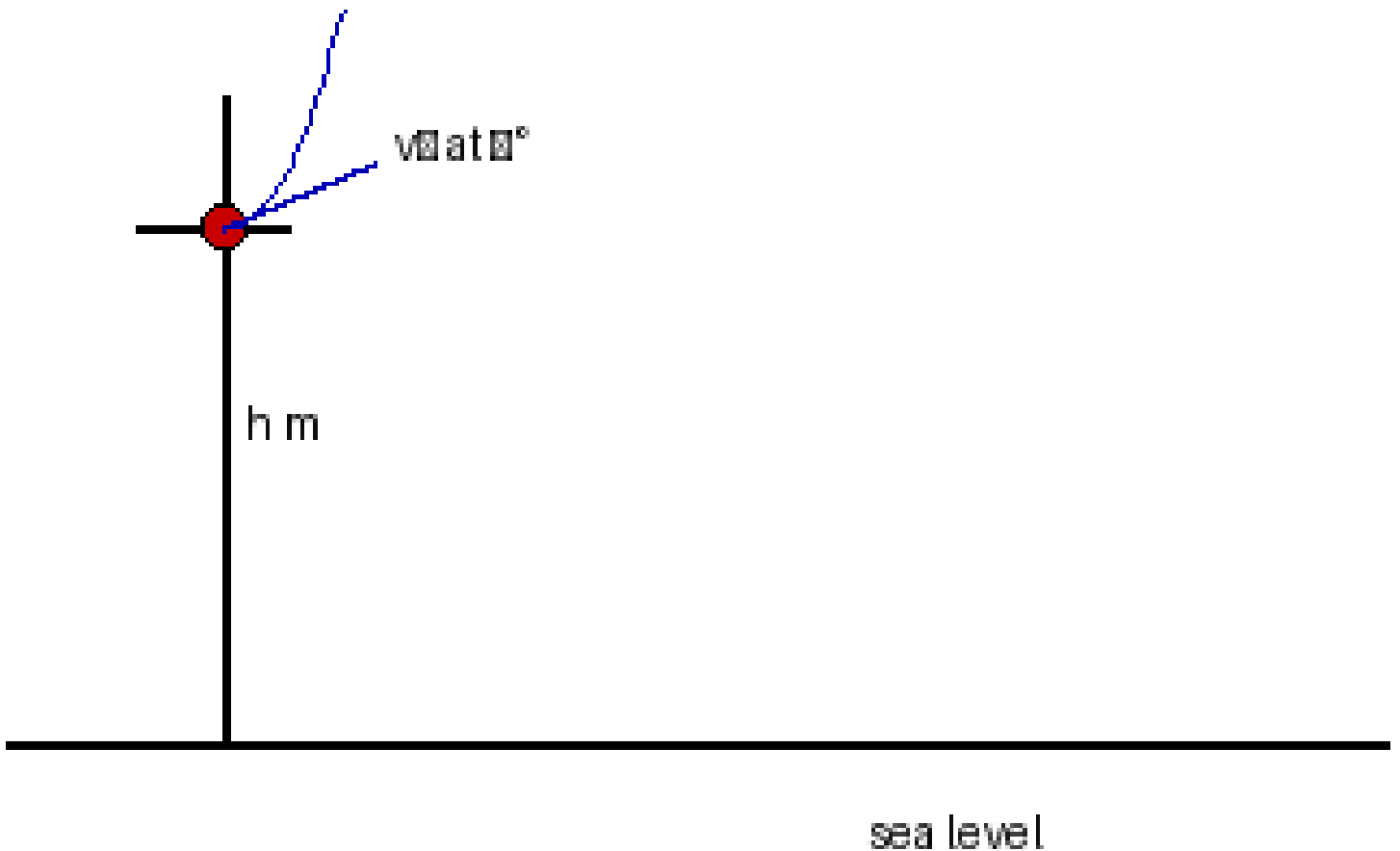
A uniform rod *AB* of mass 6 kg and length 5 m rests with end *A* against a rough vertical wall and end *B* on a smooth horizontal floor. The rod makes an angle θ with the horizontal. A particle of mass 4 kg is attached to *B*. The system is shown in the diagram below.



- By taking moments about A , show that the normal reaction at B is 52.1 N (to 3 s.f.). [4]
- Find the magnitude of the frictional force at A . [3]
- Given that the rod is on the point of slipping, find the coefficient of friction between the rod and the wall. [3]
- State, with a reason, whether the rod would begin to slip if the 4 kg particle were placed at the midpoint of AB instead of B . [2]

4. [12 marks]

A particle is projected from a point O at the top of a vertical cliff of height 78.4 m above sea level, with speed 28 m s^{-1} at an angle α above the horizontal. The particle hits the sea at point P as shown in the diagram below.



- Given that $\alpha = 20^\circ$, find the time taken for the particle to reach the sea. [3]
- Find the horizontal distance OP . [2]
- Find the speed and direction of motion of the particle when it hits the sea at P . [4]
- Find the value of α for which the horizontal distance OP is a maximum. [3]

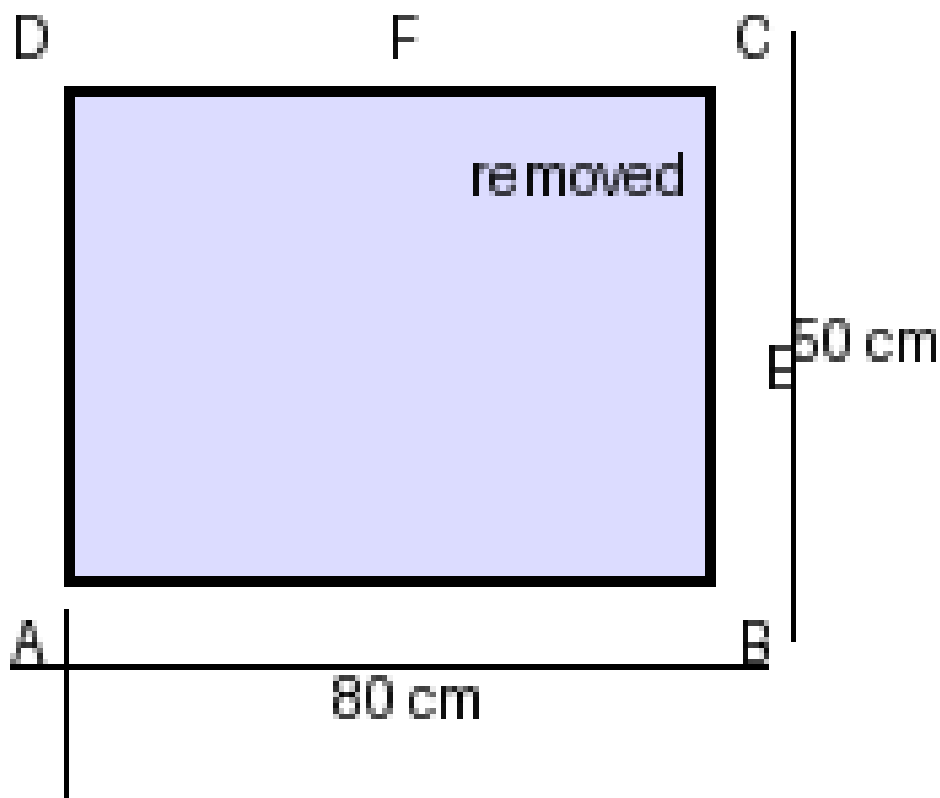
5. [9 marks]

Two particles A and B , of masses 4 kg and 6 kg respectively, are connected by a light inextensible string passing over a small smooth pulley fixed at the top of a rough plane inclined at 40° to the horizontal. Particle A lies on the plane and particle B hangs vertically below the pulley. The coefficient of friction between A and the plane is 0.3 . The string lies along a line of greatest slope of the plane. The system is released from rest.

- Find the acceleration of particle B . [4]
- Find the tension in the string. [2]
- When particle B has fallen a distance of 2 m, the string breaks. Find the total distance travelled by particle A from the moment of release until it first comes to rest. [3]

6. [10 marks]

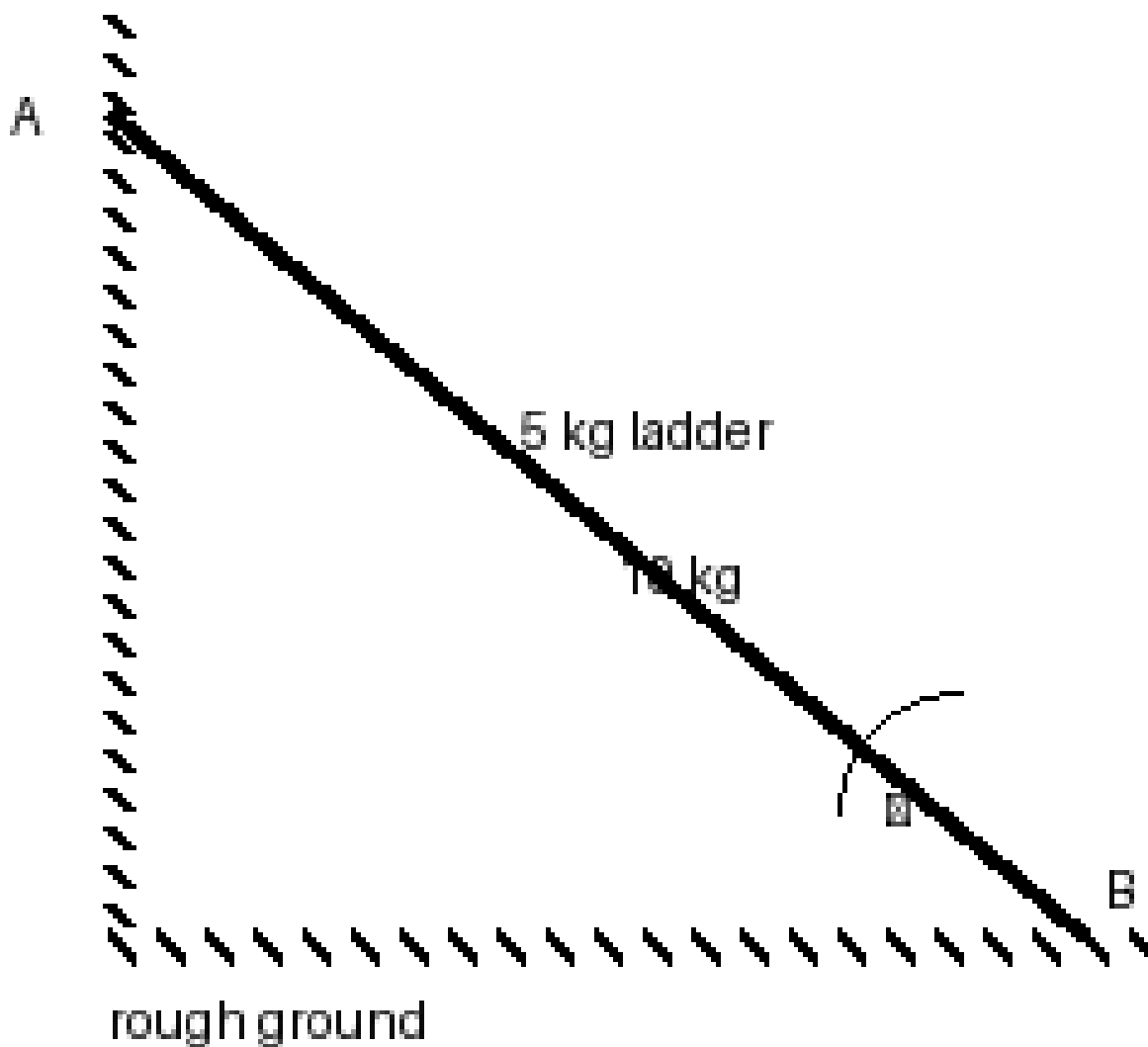
A uniform rectangular lamina $ABCD$ has $AB = 80$ cm and $AD = 50$ cm. A small triangular portion CEF is removed, where E and F are the midpoints of BC and CD respectively. The lamina is suspended from A . The diagram below shows the lamina and the removed portion.



- Show that the centre of mass of the original rectangle $ABCD$ is at the point $(40, 25)$ from A . [1]
- Find the coordinates of the centre of mass of the triangular portion CEF from A . [2]
- Hence find the coordinates of the centre of mass of the remaining lamina from A . [4]
- Find the angle that AB makes with the vertical when the lamina hangs in equilibrium from A . [3]

7. [13 marks]

A uniform ladder AB of mass 5 kg and length 4 m rests in limiting equilibrium with end A against a smooth vertical wall and end B on rough horizontal ground. A person of mass 10 kg stands at the midpoint of the ladder. The ladder makes an angle θ with the horizontal, where $\tan \theta = 2$. The set-up is shown in the diagram below.



- (a) By resolving horizontally and vertically, show that the normal reaction at A equals the frictional force at B , and that the normal reaction at B is 147 N . [3]
- (b) By taking moments about B , find the normal reaction at A . [4]
- (c) Hence find the coefficient of friction between the ladder and the ground. [3]
- (d) The person now climbs up the ladder towards A . Explain, without any further calculation, whether the ladder will begin to slip before the person reaches A . [3]

END OF PAPER